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COURSE: ITAI 1378

TIER 1

FAKE CURRENCY DETECTION

THE PROBLEM

- Counterfeit money can cause financial losses for businesses and individuals.
- Small businesses and cash-based stores often do not have advanced tools to detect fake bills.
- An automated system could help quickly verify whether a banknote is real or fake.

MY SOLUTION

- Build a computer vision model that analyzes an image of a banknote and predicts whether it is real or fake.
- The system will take an image of a banknote, process it using a trained deep learning model, and output a prediction: real or fake.

TECHNICAL APPROACH

- Technique: Image Classification
- Model: ResNet50 (transfer learning)
- Framework: PyTorch
- ResNet models perform well on image classification tasks and transfer learning allows training with smaller datasets.

DATA PLAN

- Source:
 - Fake Currency Data from Kaggle
 - <https://www.kaggle.com/mdladla/fake-currency-data>
- Size: Estimated about hundreds to thousands of images
- Labels: Real vs Fake banknotes
- Preparation: Resize images, normalize data, and split into training and testing sets.

SYSTEM DIAGRAM

- Input: Image of banknote
- Step 1: Image preprocessing (resize, normalization)
- Step 2: CNN model inference (ResNet50)
- Output: Prediction – Real or Fake banknote

SUCCESS METRICS

- Primary Metric: Classification Accuracy
- Target: $\geq 90\%$ accuracy
- Secondary Metric: Inference Speed (< 1 second per image)
- Additional Evaluation: Confusion matrix and precision/ recall.

WEEK-BY-WEEK PLAN

- Week 10: Get dataset and set up environment
- Week 11: Train initial model
- Week 12: Evaluate and improve model
- Week 13: Build demo notebook
- Week 14: Final testing and documentation
- Week 15: Final presentation

CHALLENGES & BACKUP PLANS

- Challenge: Dataset size may be small
- Plan B: Apply data augmentation

- Challenge: Model accuracy is too low
- Plan B: Try different models such as MobileNet or EfficientNet

RESOURCES NEEDED

- Compute: Google Colab
- Frameworks: PyTorch, Torchvision
- Libraries: NumPy, Matplotlib, Scikit-learn
- Estimated Cost: \$0 (using free Colab GPU)